ECB80

COUNTERBALANCE
FORKLIFT SERIES



NEW ENERGY FORKLIFT LEADER

BYD Efficient Lithium-Ion Electric Forklift

www.byd-forklifts.com

STANDARD FEATURES

Safety

- Operator Presence Sensor (OPS)
- Hill hold control
- Seat belt alarm
- Programmable automatic speed limit
- Cylinder lowering speed limit
- Charging protection
- Reversing buzzer
- LED headlights and rear combination lights
- Overload protection

Performance

- Lithium iron-phosphate battery
- AC drive system
- AC hydraulic lift system
- Battery rapid charging & opportunity charging
- Programmable performance
- Digital (LCD) multifunction display
  - Time & date
  - OPS timing display
  - Battery capacity indicator
  - Over-discharge warning (lift interrupt)
  - Parking brake indicator
  - Direction indicator
  - Maintenance indicator
  - Overload indicator

Operability & Ergonomics

- Front & rear operator assist grips
- Heavy-duty non-slip floor mat
- Tray with Cup holder

Serviceability

- AC drive motor (no brushes or commutator)
- AC hydraulic motor (no brushes or commutator)
- Charging port analyzing connection
- Easy-access no-tools floorboard
- Maintenance-free battery pack
- Side-out battery hood
- Hydraulic fluid level check
- Easy-access controllers
- CAN-bus controller wiring

Body

- 55'' load backrest
- Drawbar pin
- Punched plate entry/exit step
- Pneumatic tires (black)
- Rearview mirrors (left & right sides)
- Rear assist grip with horn button
- Orange seatbelt
- Retractable seatbelt
- Overhead guard

OPTIONAL EQUIPMENT

Load Handling (Wide visible mast)

- 2-stage standard mast (Simplex)
- 3-stage single Full free-lift mast（Triplex FFL)
- Customized fork
- Customized load backrest

Load Handling (Hose & Valve Assembly)

- 4-way valve (no 3rd & 4th function hosing)
- 4-way valve (with 3rd function hosing)
- 4-way valve (with 3rd & 4th function hosing)
- Hydraulic DN10 connection on 3rd lever
- Hydraulic DN10 connection on 4th lever
- Automatic fork-leveling control
- Hang-on side shifter
- Fork positioners
- Clamps

Lights & Indicators

- Strobe light
- LED side-mounted warning laser light
- LED warning spot light
- Rear working light

Body

- Customized painting
- Fork camera
- Reversing radar
- Reversing camera
- Driver cooling fan
- Synchronous steering
- Mini lever
- Mini dual joystick
- Flat internal rearview mirror
- Panoramic rearview mirror
- Full-sealing carbin
- Retractable seatbelt
- Anti-static strap
- Special key
- Fork sleeves
- Solid tires
- Non-marking tires

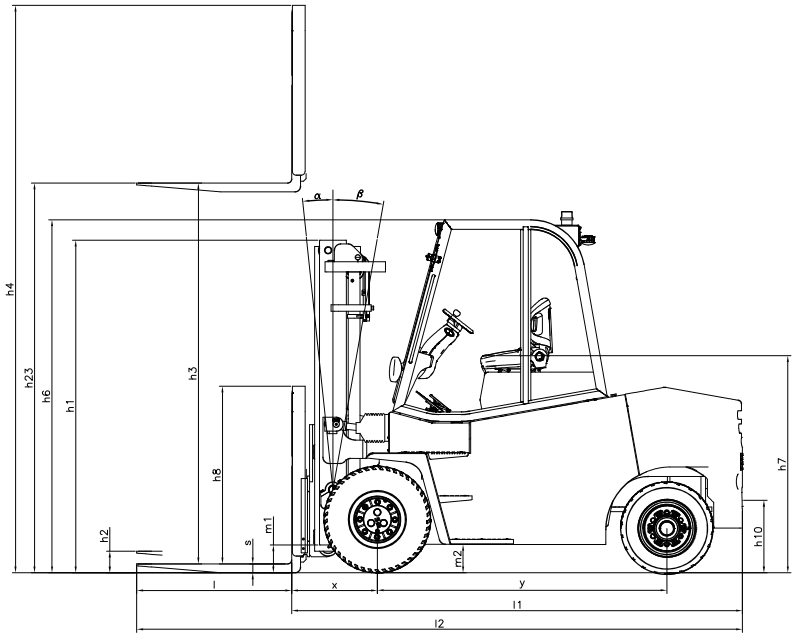
COUNTERBALANCE FORKLIFT SPECIFICATIONS

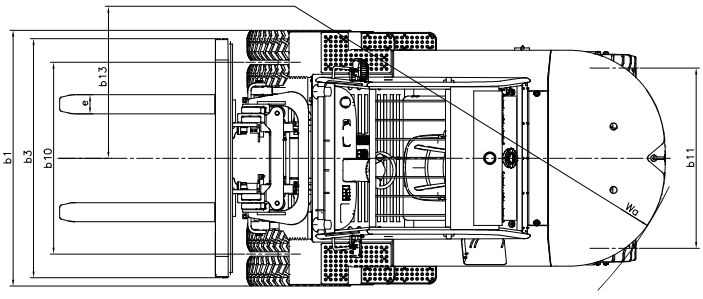
Characteristic	Manufacturer				BYD	
	Model				ECB80B/ECB80C	
	Power Type				Electric	
	Operation Type				Seat	
	Load Capacity	Q	lbs	kg	17637	8000
	Load Center	c	in	mm	24	600
	Axle Center To Fork Face	x	in	mm	27	675
	Wheel Base	y	in	mm	90	2280
Weights	Service Weight		lbs	kg	29321	13300
	Axle Load with Load, Front/Rear		lbs	kg	12963/16358	5880/7420
	Axle Load without Load, Front/Rear		lbs	kg	40454/6504	18350/2950
Wheels	Tire Type				Pneumatic	
	Tire Size, Front				8.25-15	
	Tire Size, Rear				250-15	
	Wheels, Number Front/Rear (x = driven)				4x/2	
	Track Width, Front	b10	in	mm	61	1541
	Track Width, Rear	b11	in	mm	57	1450
Dimensions	Tilt Range (forward/backward)	α/β	deg	deg	6/10	
	Height of Mast, Lowered	h1	in	mm	103	2620
	Free Lift (Top of Forks)	h2	in	mm	7	170
	Maximum Fork Height	h3	in	mm	118	3000
	Height of Mast, Extended	h4	in	mm	176	4470
	Height of Overhead Guard (Cabin)	h6	in	mm	109	2780
	Height of Seat	h7	in	mm	67	1708
	Towing Coupling Height	h10	in	mm	22	570
	Length to Fork Face	l1	in	mm	140	3550
	Overall Length	l2	in	mm	188	4770
	Overall Width	b1	in	mm	82	2076
	Fork Dimensions	s/e/l	in	mm	3/6/48	70/150/1220
	Mounting Class				4A	
	Width of Load Backrest	b3	in	mm	71	1800
	Ground Clearance, Below Mast	m1	in	mm	9	220
	Ground Clearance, Centre of Wheelbase	m2	in	mm	9	220
	Aisle Width with Pallet 1000 x 1200 Across Forks	Ast	in	mm	211	5355
	Aisle Width with Pallet 800 x 1200 Along Forks	Ast	in	mm	219	5555
	Turning Radius	Wa	in	mm	128	3260
	Turning Radius, Inside	b13	in	mm	47	1200
Performance	Travel Speed, with/without Load		mph	Km/h	10/11	16/17
	Lifting Speed, with/without Load		ft/min	mm/s	79/69	400/350
	Lowering Speed, with/without Load		ft/min	mm/s	98/79	500/400
	Tractive Force, with/without Load		KN	KN	32/28	
	Gradeability, with/without Load		%	%	15/20	
	Service/Parking Brake				Hydraulic	
Drive	Drive Motor, 60 Minute Rating		HP	KW	18.8x2	14x2
	Lift Motor, Rating at S3 15%		HP	KW	32.2x2	24x2
	Battery Voltage/Rated Capacity		V/Ah	V/Ah	80/1080	
	Battery Weight		lbs	kg	2646	1200
Others	Type of Drive Control				AC	
	Operating Pressure for Attachments		psi	bar	18	
	Noise Level at Operator's Ear		dB(A)	dB(A)	≤80	

MAST SPECIFICATIONS

ECB80 (WITH FORK ONLY)	Maximum Fork Height (h3)	Free Lift Height with 1200mm Load Backrest (h2)	Lowered Height (h1)	Extended Height with 1200mm Load Backrest (h4)	Tilt Range, FWD(α)	Tilt Range, BWD(β)	Load Capacity at 24 in. (610mm) Load Center with Standard Tread
	in (mm)	in (mm)	in (mm)	in (mm)	deg	deg	lbs (kg)
2-Stage Mast (Duplex STD)	118 (3000)	7 (170)	103 (2620)	176 (4470)	6°	10°	17637(8000)
	130 (3300)	7 (170)	109 (2770)	188 (4770)	6°	10°	17637(8000)
	138 (3500)	7 (170)	113 (2870)	196 (4970)	6°	10°	/
	157 (4000)	7 (170)	123 (3120)	215 (5470)	6°	10°	/
	177 (4500)	7 (170)	133 (3370)	235 (5970)	6°	6°	/
	197 (5000)	7 (170)	143 (3620)	255 (6470)	6°	6°	/
	217 (5500)	7 (170)	152 (3870)	274 (6970)	6°	6°	/
	236 (6000)	7 (170)	162 (4120)	294 (7470)	6°	6°	/
3-Stage Single Full Free-lift Mast (Triplex FFL)	189 (4800)	59 (1487)	107 (2720)	247 (6270)	6°	6°	/
	197 (5000)	61 (1557)	110 (2790)	255 (6470)	6°	6°	/
	217 (5500)	68 (1717)	116 (2950)	274 (6970)	6°	6°	/
	236 (6000)	76 (1937)	125 (3170)	294 (7470)	6°	6°	/
	256 (6500)	83 (2107)	131 (3340)	314 (7970)	6°	3°	/
	276 (7000)	91 (2317)	140 (3550)	333 (8470)	6°	3°	/
	295 (7500)	98 (2487)	146 (3720)	353 (8970)	6°	3°	/

LAYOUTS





BYD Forklift production site is certified:

ISO 9001 ISO 14001 OHSAS 18001